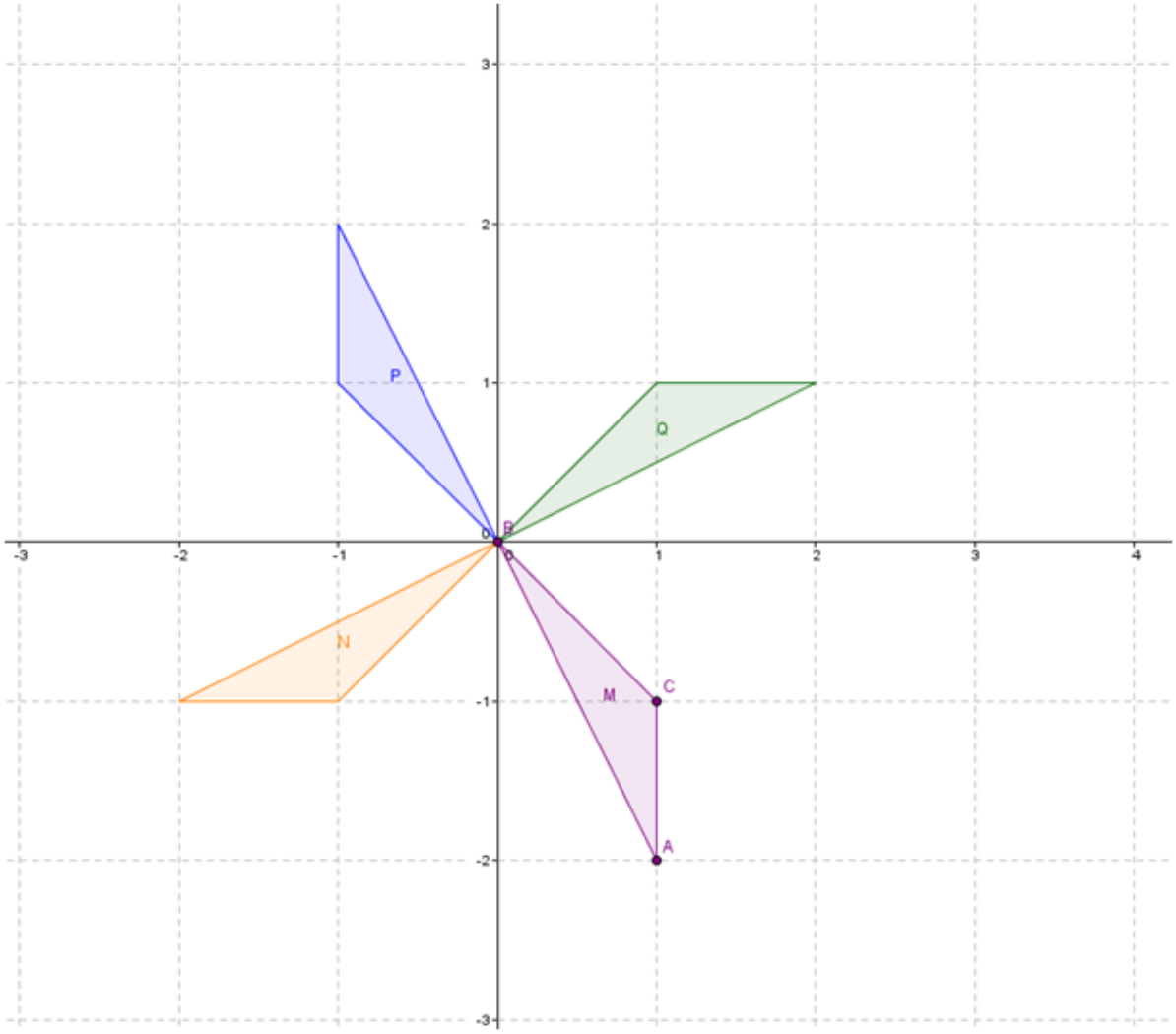
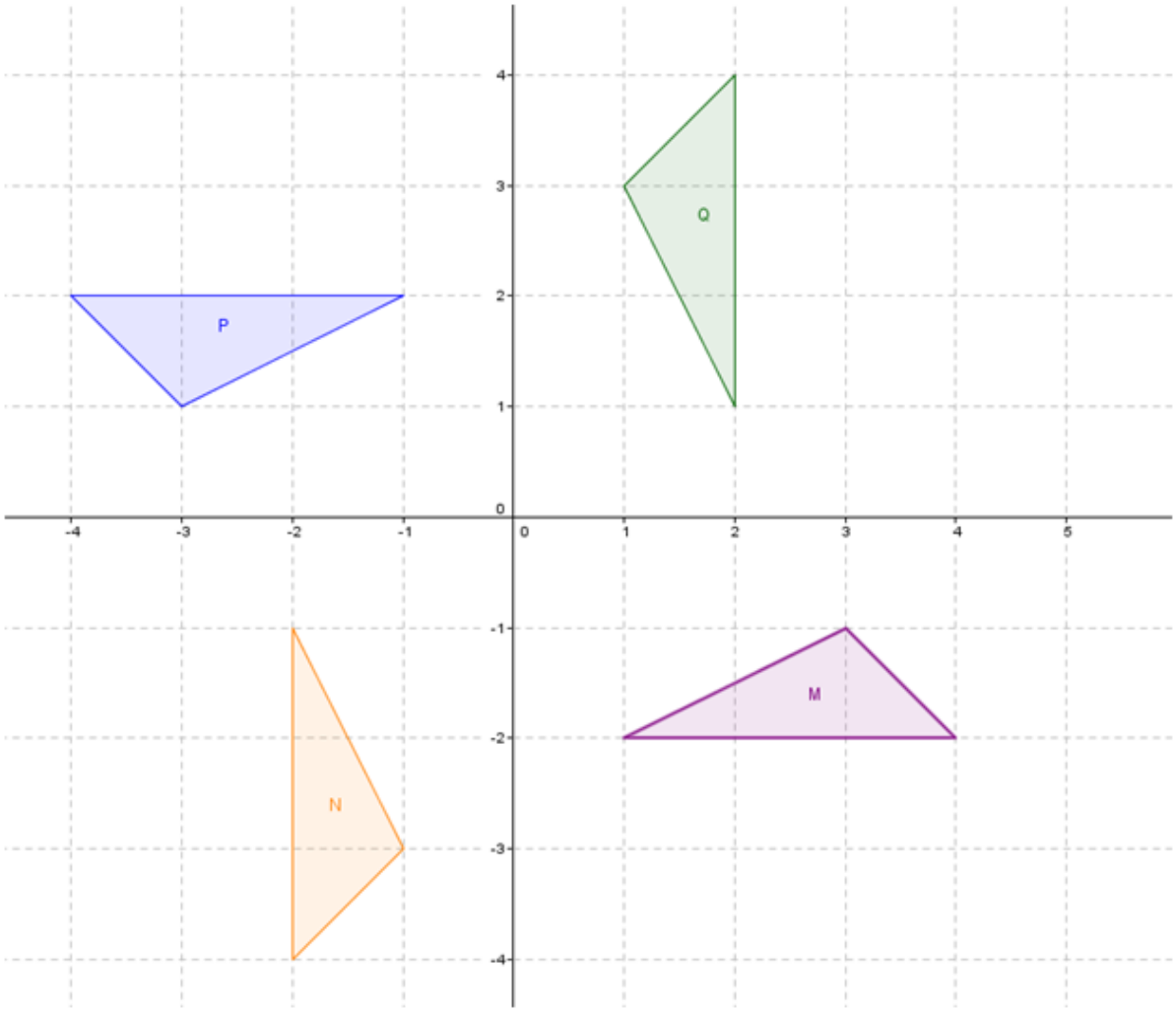
Rotations

1 Consider the diagram shown.
Identify the correct image after M is rotated 180° clockwise about the origin.



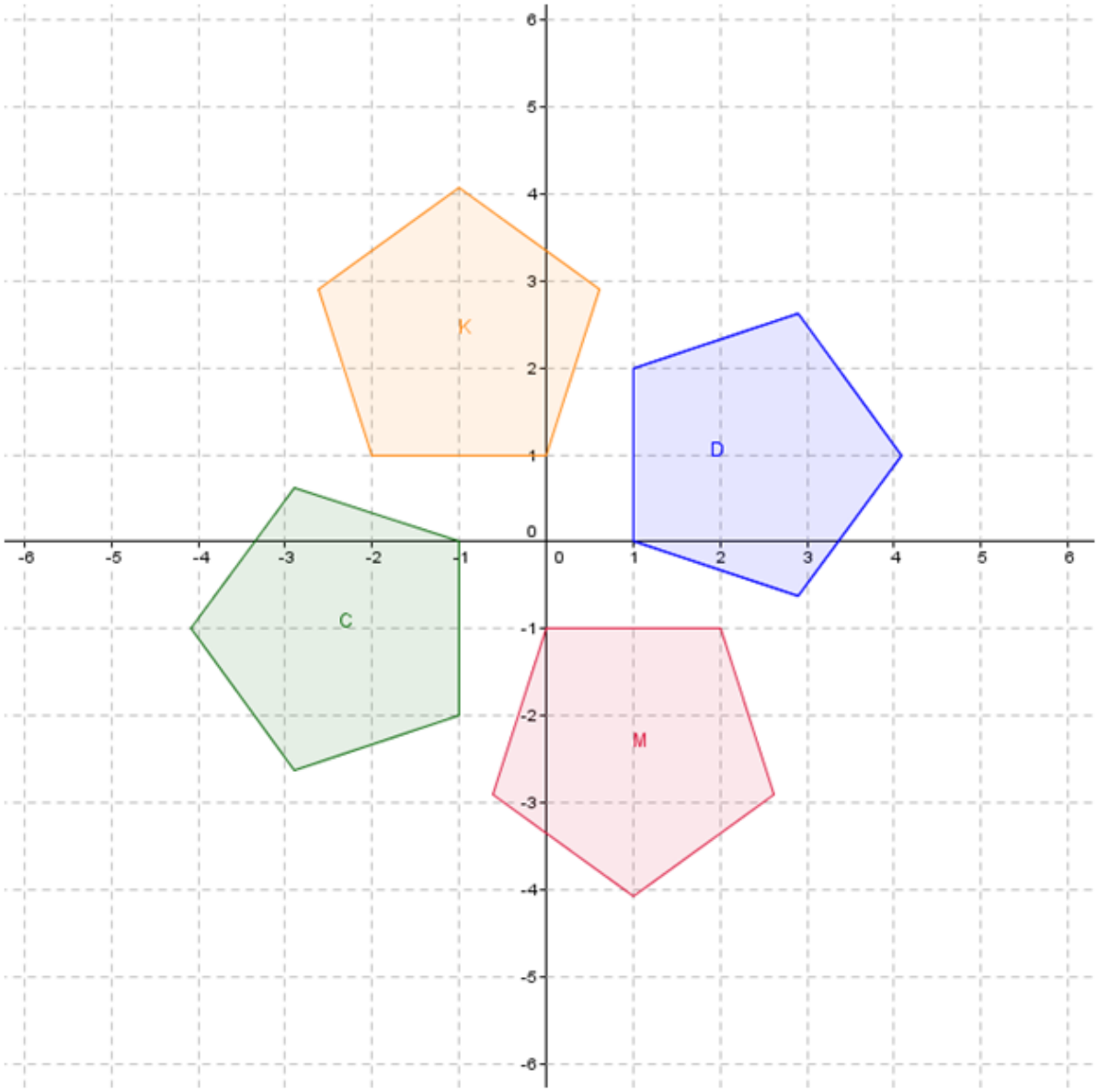
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2 Consider the diagram shown.
Identify the correct image after Q is rotated 270° clockwise about the origin.



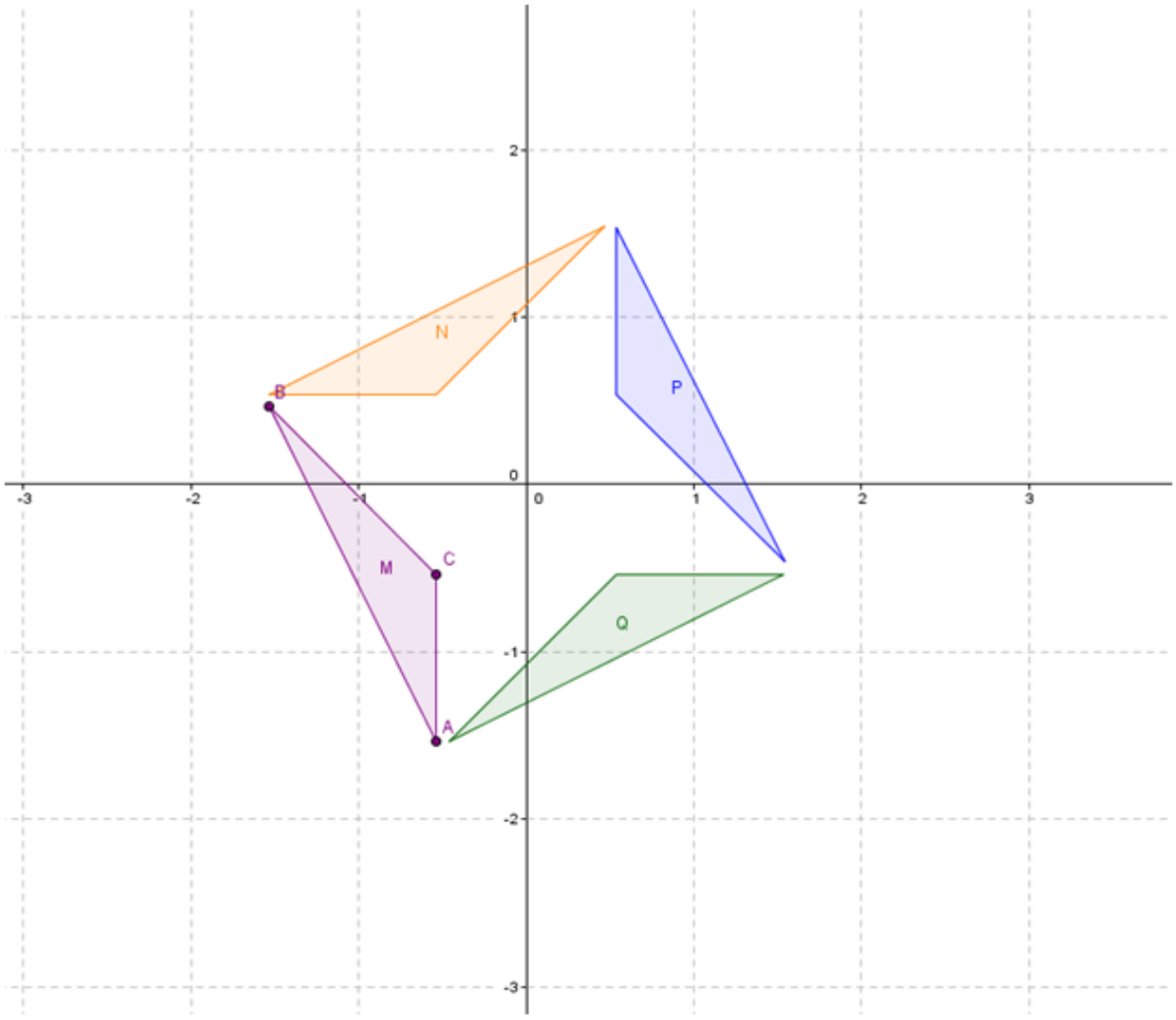
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3 Consider the diagram shown.
Identify the correct image after M is rotated 90° counterclockwise about the origin.



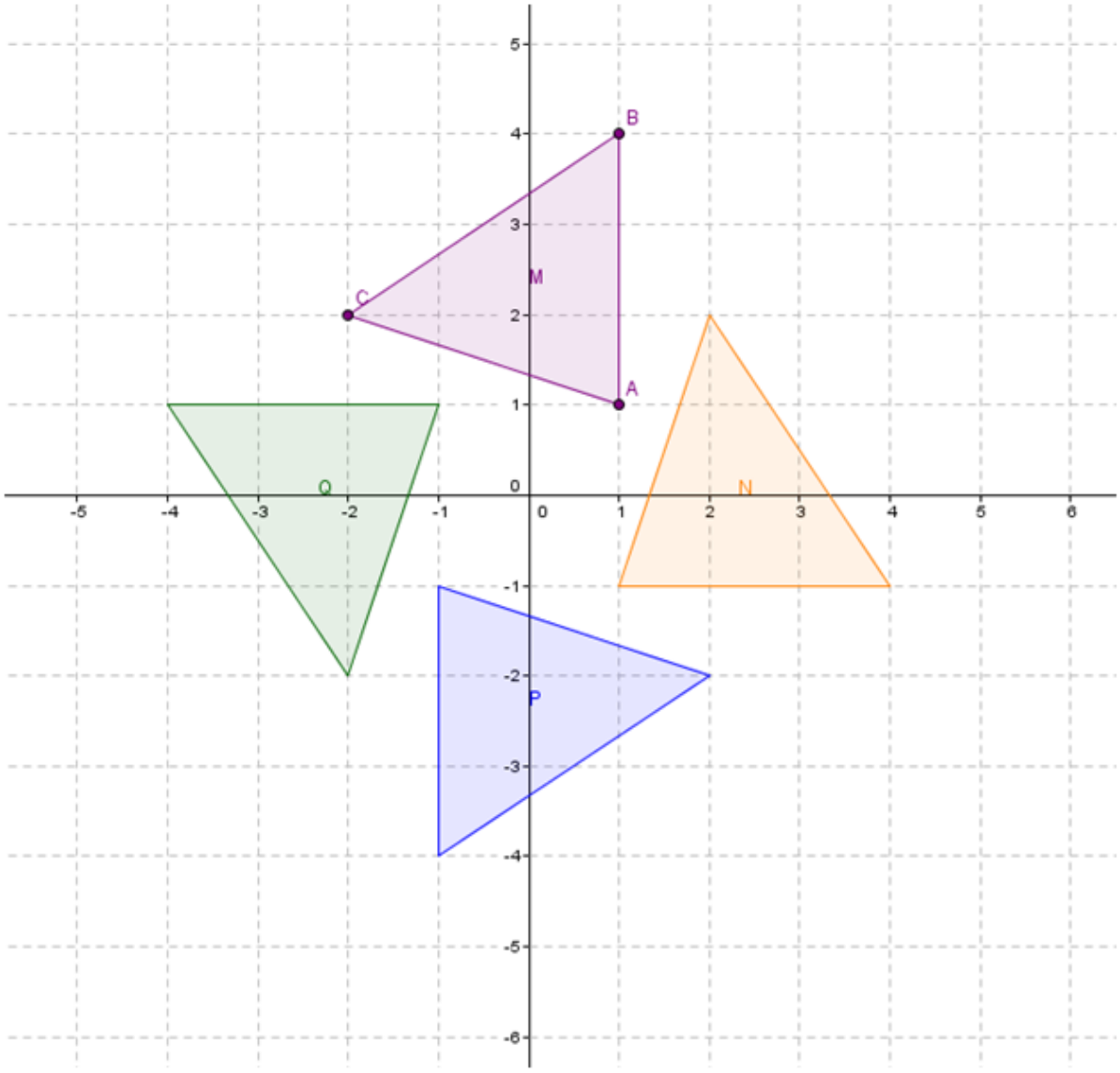
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4 Consider the diagram shown.
Identify the correct image after Q is rotated 270° counterclockwise about the origin.



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5 Consider the diagram shown.
Identify the correct image after Q is rotated 90° counterclockwise about the origin.



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8 For each of the following sets of points:

i Plot the given A, B, C, and D on the same coordinate plane.

ii Plot the points A', B', C', and D' that would result when we rotate the original points A, B, C, and D by 90° clockwise about the origin.

a $A\,(5,5)$, $B\,(9,5)$, $C\,(9,9)$, and $D\,(5,9)$

b $A\,(-5,5)$, $B\,(0,5)$, $C\,(0,7)$, and $D\,(-5,7)$

9 For each of the following sets of points:

i Plot the given A, B, C, and D on the same coordinate plane.

ii Plot the points A', B', C', and D' that would result when we rotate the original points A, B, C, and D by 90° counterclockwise about the origin.

a $A\,(1,4)$, $B\,(4,4)$, $C\,(4,7)$, and $D\,(1,7)$

b $A\,(-5,-5)$, $B\,(-1,-5)$, $C\,(-1,-1)$, and $D\,(-5,-1)$

10 Consider the following.

a Plot the point $A\,(6,0)$.

b A is rotated 90° clockwise about the origin to produce B.
Determine the coordinates of B.

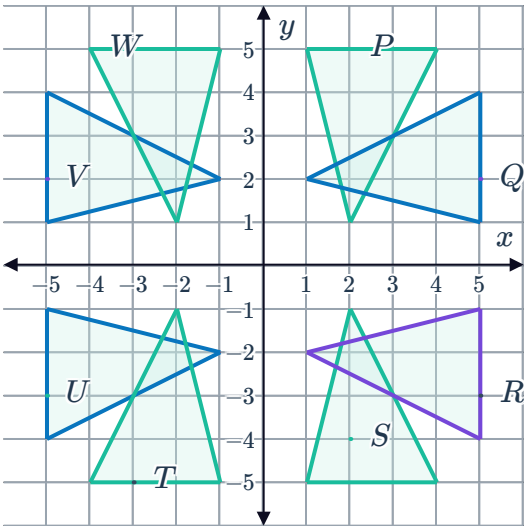
c Describe the two translations that can be used to move point A to point B.

Additional questions

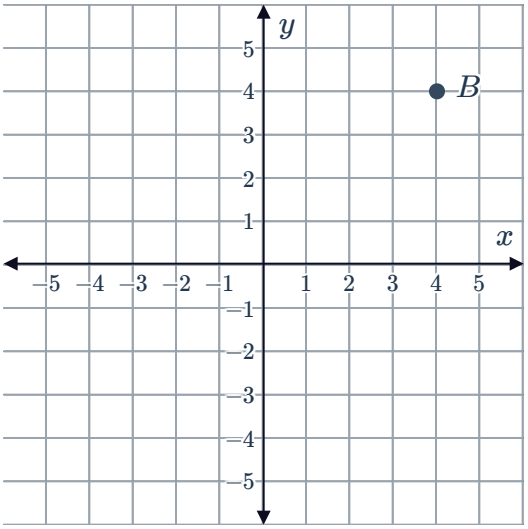
11 Describe the result of rotating a figure 90° clockwise about the origin.

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12 If $\triangle R$ is rotated 270° clockwise about the origin, which triangle is the resulting image?

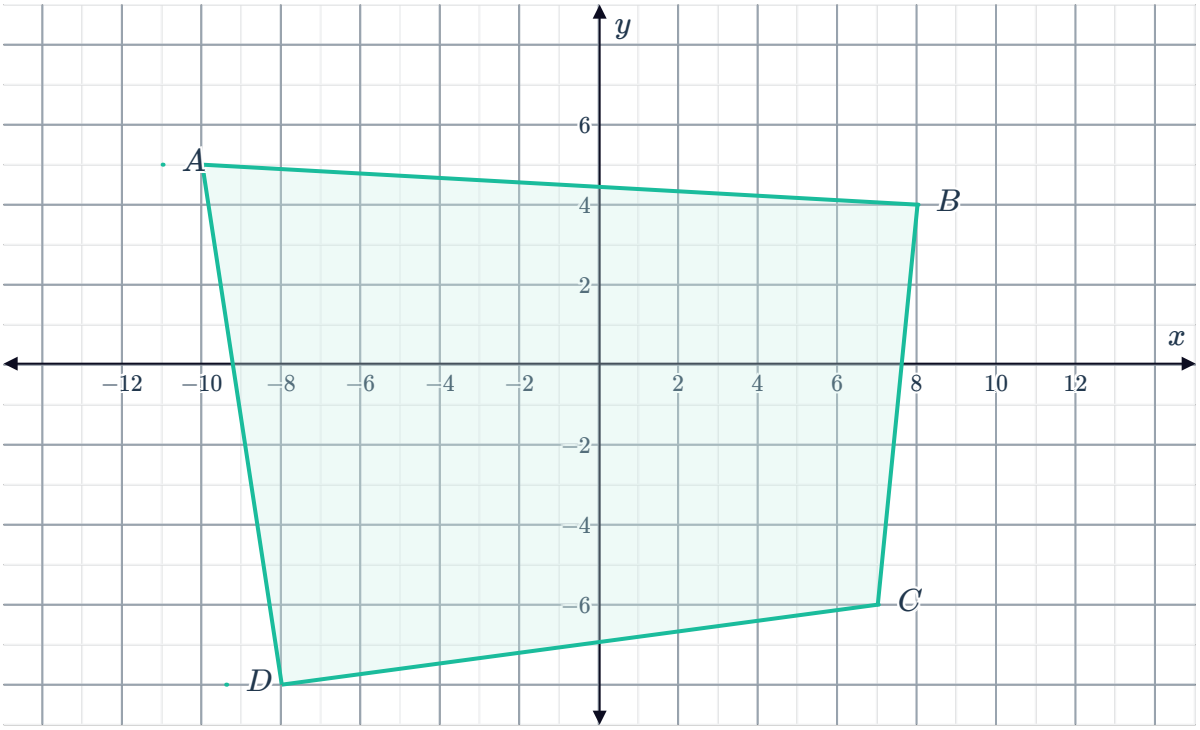


13 What are the coordinates of B', the image produced by rotating B 90° clockwise about the origin?



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14 Rotate $ABCD$ 180° counterclockwise about the origin.

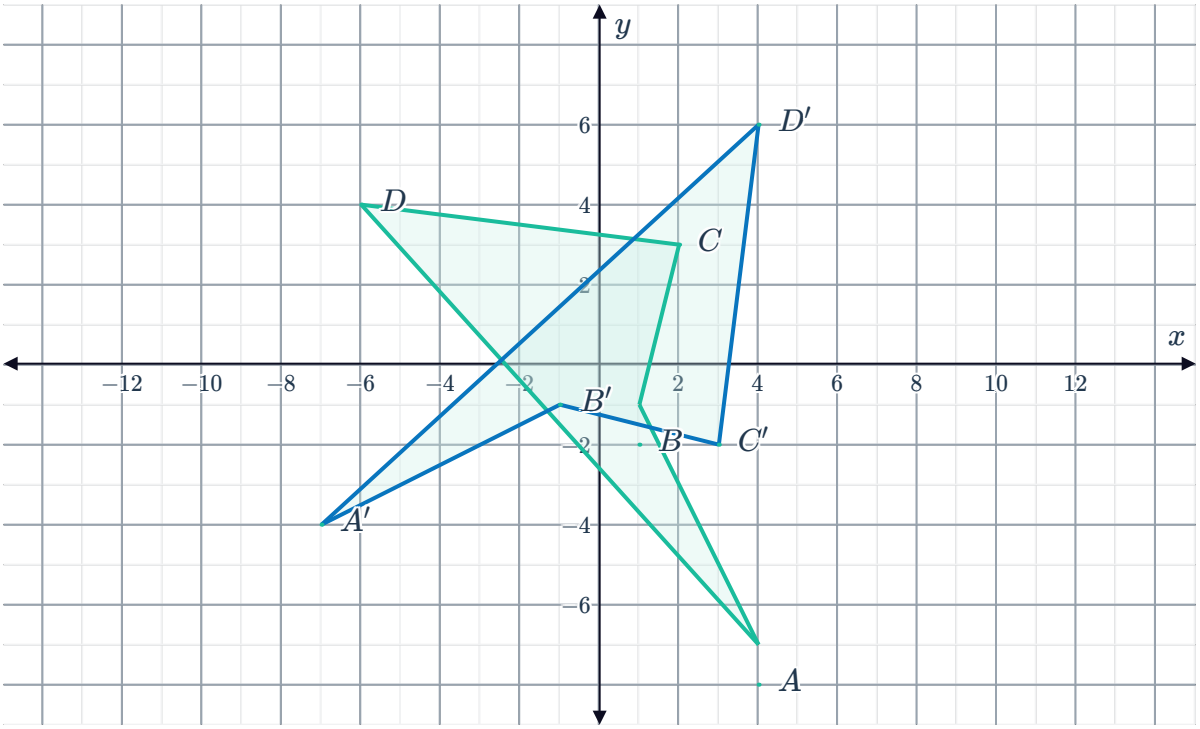


15 The vertices of $\triangle ABC$ are located at $A(1,1)$, $B(4,1)$, and $C(2,-3)$.

a Graph $\triangle ABC$ on the coordinate plane.

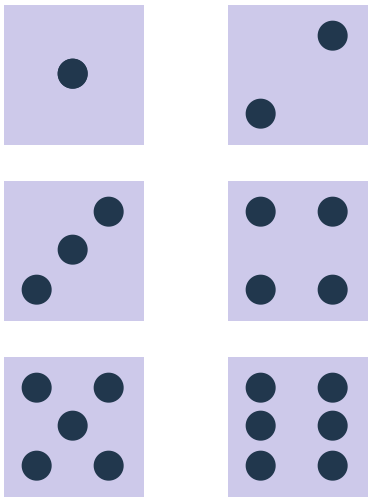
b Rotate $\triangle ABC$ 270° clockwise about C. Label the vertices and give their coordinates.

16 Describe the rotation required to map $ABCD \rightarrow A'B'C'D'$.



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17 Akilah was looking at the faces on a die. She noticed that the face with a single dot looks the same when rotated by 180°.



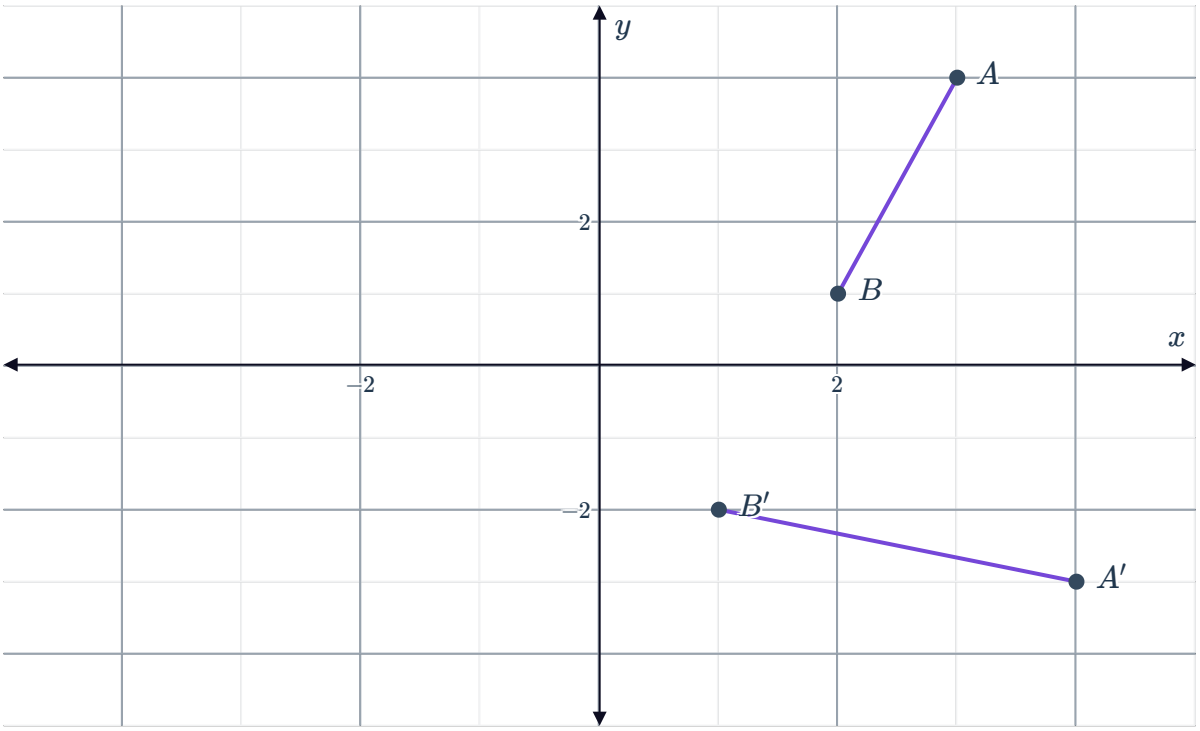
a Akilah hypothesized that each face looks the same when rotated 180°. Is she correct? Explain your reasoning.

b Akilah also hypothesized that each face looks the same when rotated 90° in either direction. Is she correct? Explain your reasoning.

18 Avery is on the corner of Argyle street and Campbell street. Argyle st. runs north-south and Campbell st. runs east-west. He has a map with a compass at the top depicting that the top of the map is north.
If Avery is facing south on Argyle street, how many degrees and in what direction does Avery need to turn the map so the top of the map is aligned with the direction he is facing? Explain how you know.

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19 Aneta has rotated the line segment $\overline{AB}$ 90° counterclockwise about the origin as shown below:



Has Aneta made an error? Explain how you know.

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